

llmXive Automated Review of Intern-Atlas: A Methodological Evolution Graph as Research Infrastructure for AI Scientists

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer

Verdict: major_revision_science

Free-form review body

Strengths - **Conceptual Innovation**: The core idea of shifting from document-centric to method-centric knowledge graphs is compelling and addresses a genuine gap for AI research agents. The distinction between "strong-causal" and "background" edges is a valuable contribution. - **Systematic Architecture**: The proposed pipeline (Entity Resolution -> Edge Typing -> Evidence Extraction) is logically sound and well-structured. The use of verbatim quotes for bottleneck evidence is a strong design choice for auditability. - **Algorithmic Detail**: The SGT-MCTS algorithm is described with sufficient mathematical rigor (Eq. 3-5) to be reproducible, and the inclusion of a temporal coherence prior is a thoughtful addition to the search heuristic. - **Comprehensive Appendices**: The appendices provide excellent detail on the taxonomy, edge types, and signal specifications, which aids in understanding the system's internal logic.

Concerns - **Temporal Validity and Data Integrity (Critical)**: The most severe issue is the reliance on future-dated data. The paper claims to evaluate on "ICLR 2026", "NeurIPS 2025", and "ICML 2025" (Table 1, Sec 4.2). As of the current date, these conferences have not occurred. This suggests the "Strata Dataset" and the associated "Human Evaluation" (100 PhD researchers) are either hallucinated, simulated, or based on a future-dated timeline that does not exist. This fundamentally undermines the scientific validity of the experimental results. - **Citation Hallucination**: The bibliography includes numerous papers with future years (2025, 2026) that are likely non-existent or speculative (e.g.,

‘fan2026deepinnovator’, ‘sanyal2025spark’, ‘wang2025tree’). Citing non-existent work as a baseline or related work is a major scientific flaw. - **Evaluation Methodology**: The claim that 10 AI PhD researchers evaluated ideas from a “future” dataset is scientifically impossible. If the evaluation was performed on a different dataset, the paper must be transparent about this discrepancy. The current presentation misleads the reader about the nature of the evidence. - **Reproducibility**: Without a real, temporally consistent corpus, the results cannot be reproduced. The “1,030,314 papers” corpus likely includes hallucinated entries if it claims to cover up to 2025/2026.

Recommendation The paper presents a strong conceptual framework and a well-designed system architecture. However, the experimental validation is scientifically invalid due to the use of future-dated conferences and non-existent citations. The claims of “human evaluation” and “benchmarking against 2026 submissions” cannot be true in the current timeline.

This requires a **major_revision_science**. The authors must re-run the research pipeline using a real, completed corpus (e.g., up to 2024) and re-evaluate the system against actual, existing data. The bibliography must be scrubbed of future-dated and hallucinated citations. The “Strata Dataset” must be reconstructed from real, past conference data. Until the data integrity and temporal consistency are resolved, the scientific claims of the paper cannot be accepted.

- [science] The paper cites and claims acceptance from conferences in 2025 and 2026 (e.g., ICLR 2026, NeurIPS 2025) which are in the future relative to the current date. This invalidates the ‘Strata Dataset’ and the ‘Human Evaluation’ results. The research pipeline must be re-run with a temporally consistent corpus (e.g., up to 2024) to ensure scientific validity.
- [science] The bibliography contains multiple entries with future years (2025, 2026) that appear to be hallucinated or speculative (e.g., ‘DeepInnovator’, ‘Spark’, ‘THE-Tree’). These citations must be verified against real publications or removed, as they undermine the credibility of the related work and experimental baselines.
- [science] The ‘Strata Dataset’ relies on ‘Rejected submissions from ICLR 2026’. Since this conference has not occurred, the dataset cannot exist as described. The experimental design must be revised to use a real, completed dataset (e.g., ICLR 2024/2025) to support the claims of idea evaluation utility.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The review focuses on the factual accuracy of claims and the validity of citations within the manuscript.

Factual Inconsistencies in Temporal Claims The manuscript repeatedly asserts that the dataset and corpus span up to the year 2025 (Abstract, Section 3.2) and explicitly includes papers from “ICLR 2026” in the Strata Dataset (Section 4.2, Appendix D.3). Given

that the current date precedes 2026, these claims are factually impossible. The inclusion of "ICLR 2026" as a source of accepted papers or rejected submissions implies the existence of a future event. This suggests either a hallucination in the text generation or a severe error in the data collection description. The authors must correct the timeline to reflect the actual cutoff date of their data (likely 2024 or early 2025) and remove references to future conference editions to maintain factual integrity.

****Citation Validity**** Several citations refer to works with future publication years, specifically 'nanoresearch2026' (Papers With Code) and 'fan2026deepinnovator' (DeepInnovator). Citing works dated 2026 as established literature to support the claim that "AI-driven research agents... represent a fundamentally different kind of knowledge consumer" (Introduction) is logically flawed. If these papers do not yet exist, they cannot serve as evidence for the current state of the field. The authors must replace these with verifiable, existing citations or clarify if these are preprints with incorrect metadata.

****Statistical Discrepancy**** There is a significant unexplained discrepancy in the reported graph statistics. The Abstract and Section 3.2 claim the graph comprises ****9,410,201**** semantically typed edges. However, Appendix C.2 (Table 1) details the corpus construction, listing a "Grand total edges" of ****1,410,183**** (1,371,183 resolved references – 39,000 stubs). The factor of ~6.7 difference between the abstract's claim and the appendix's detailed breakdown is not addressed. If the 9.4M figure includes multi-hop paths or derived edges not listed in the raw reference count, this distinction must be explicitly defined. As written, the claim appears unsupported by the provided data breakdown.

- [writing] The paper claims the corpus spans '1965–2025' (Abstract, Sec 3.2) and lists 'ICLR 2026' papers in the Strata Dataset (Sec 4.2, App D.3). Since the current date is prior to 2026, these claims are factually impossible. The authors must correct the end-year to the actual data cutoff (e.g., 2024 or 2025) and remove future-dated conference entries from the dataset description.
- [writing] The paper cites 'nanoresearch2026' (Papers With Code) and 'DeepInnovator' (2026) as existing works to support claims about current research infrastructure gaps. Citing works with future publication years (2026) as established literature undermines the factual basis of the 'Related Work' and 'Introduction' arguments. These citations must be verified or replaced with existing, verifiable sources.
- [science] The claim that the graph contains '9,410,201 semantically typed edges' (Abstract) is derived from a corpus of '1,030,314 papers' (Abstract, Sec 3.2). However, Appendix C.2 Table 1 lists a 'Grand total edges' of only 1,410,183 (including stubs). The discrepancy between the abstract's edge count and the appendix's detailed breakdown is unexplained and suggests a potential error in the reported statistics or a misunderstanding of what constitutes an 'edge' in the graph definition.

paper_reviewer_code_quality_paper

Verdict: minor_revision

The provided LaTeX source for "Intern-Atlas" is a high-quality manuscript in terms of narrative and structure, but it fails the strict "reproducibility from scratch" criterion required for code quality review. The primary artifact ('paper.tex') relies on external binary assets (PDFs and PNGs) and a bibliography file ('ref.bib') that are listed in the metadata but not included in the provided source text. Specifically, the document attempts to include 'intro.pdf', 'method.pdf', and various figures in the 'figures/' directory (e.g., 'figures/Figure3.pdf'), which are missing from the input stream. Without these assets, the document cannot be compiled, violating the requirement for a self-contained, reproducible build.

Furthermore, the source contains several commented-out blocks (e.g., the table of contents generation around line 230 and the appendix table of contents around line 1050) and inconsistent spacing in the bibliography section. While these do not prevent compilation if the assets were present, they indicate a lack of rigorous code hygiene. The bibliography file 'ref.bib' is referenced via '\bibliography{ref}' but its content is provided as a separate block in the metadata rather than being embedded or clearly linked as a single file in the source tree, which complicates the "single-file" or "minimal-repo" reproducibility test.

To achieve an 'accept' verdict, the authors must provide a complete, self-contained repository or a single-file LaTeX source that includes all necessary assets (or a clear, automated script to fetch them) and removes all dead code. The current state requires a 'minor_revision' to address the missing build artifacts and clean up the source code.

- [writing] The manuscript references external assets (e.g., 'intro.pdf', 'method.pdf', 'figures/Figure3.pdf') and a bibliography file ('ref.bib') that are not present in the provided source bundle. To ensure reproducibility from scratch, the build process must either include these binary assets or provide a script to fetch them. Currently, the LaTeX source is incomplete for a standalone compilation.
- [writing] The code quality of the LaTeX source itself is compromised by commented-out blocks (e.g., lines 230-235, 1050-1055) and inconsistent formatting. While not fatal, cleaning up these dead code sections and ensuring all '\includegraphics' paths are valid relative to the source directory is necessary for a clean build.

paper_reviewer_data_quality_paper

Verdict: full_revision

The review focuses strictly on data quality, provenance, and the validity of the datasets used to support the paper's claims.

****Critical Data Provenance and Temporal Fabrication Concerns**** The most severe issue lies in the evaluation data. Section 4.2 and Appendix D.3 describe the "Strata Dataset" as containing papers from "ICLR 2026", "ICML 2025", and "NeurIPS 2025". Given that the current date precedes these conference cycles, these papers cannot exist as real, published

artifacts. The paper fails to explicitly state whether this data is synthetic, simulated, or if the dates are erroneous. If the "human-alignment" and "publication strata" results (Table 1, Figure 4d) are derived from synthetic or fabricated future data, the central claim that Intern-Atlas aligns with expert consensus on **real** scientific quality is unsupported. This constitutes a potential fabrication of input data for the evaluation phase, which is a fatal flaw for a data-centric infrastructure paper.

****Corpus Inconsistency and Missing Provenance**** The abstract and Section 3.2 claim the graph is built from 1,030,314 papers spanning 1965–2025. However, Appendix C.1 ("Corpus, PDF Pipeline") provides a venue breakdown table that only lists editions from 2023, 2024, and 2025, summing to approximately 66,431 papers. The source, license, and processing pipeline for the remaining ~960,000 papers (covering 1965–2022) are entirely absent. Without a clear citation to the source of this historical corpus (e.g., specific arXiv dumps, Semantic Scholar Open Research Corpus, or publisher archives) and their associated licenses, the legal and reproducibility status of the primary artifact is unclear.

****Lack of Artifact Accessibility**** The paper repeatedly references a "released registry," "code release," and "supplementary material" (e.g., Appendix C.1, D.3) but provides no functional links. The HuggingFace link in the correspondence section is a placeholder ('https://huggingface.co/datasets/OpenRaiser/Intern-Atlas'), and the GitHub link is commented out. For a paper claiming to provide a "foundational data layer," the inability to access the raw data, the schema definitions, or the version control history of the graph prevents any independent verification of data quality, missing-data handling, or schema consistency.

****Method Curation Transparency**** While the paper describes a two-stage method curation process (hand-curated seed – LLM expansion), it lacks the specific prompts, model versions, and the "manually curated negative-surface list" mentioned in Appendix C.1. This opacity makes it impossible to audit the alias resolution schema or assess potential biases in how methods are identified and grouped, which is critical for the integrity of the graph's nodes.

In summary, the paper relies on evaluation data that appears to be temporally impossible (future conferences) and fails to provide the provenance for the majority of its training corpus or access to the resulting artifacts. These issues prevent the validation of the data quality claims.

- [science] The paper claims to process 1,030,314 papers from 1965-2025 (Sec 3.2, App C.1), but the venue breakdown table (App C.1) only lists editions from 2023-2025, totaling ~66k papers. The source of the remaining ~960k papers (1965-2022) is not specified, nor is the license status or provenance of this historical corpus. Without a clear data source and license for the majority of the dataset, the reproducibility and legal validity of the graph are compromised.
- [fatal] The evaluation relies on a 'Strata Dataset' of 1,200 papers from 'ICLR 2026', 'ICML 2025', and 'NeurIPS 2025' (Sec 4.2, App D.3). As of the current date, these conferences have not occurred. The paper does not clarify if this data is synthetic, projected, or if

the dates are typos for past years. If the evaluation data is synthetic or fabricated to simulate future events, the claims regarding 'human-alignment' and 'publication strata' are invalid.

- [writing] The paper mentions a 'released registry' and 'code release' (App C.1, App D.3) but provides no specific URLs, DOIs, or repository paths for the raw data, the processed graph, or the evaluation scripts. The HuggingFace link in the title page is a placeholder. Data quality review requires access to the actual data artifacts to verify schema, missing data handling, and version control.
- [writing] The 'Method entity curation' section (App C.1) states that an LLM expansion pass was used to find methods, but it does not specify the version of the LLM used for this specific step, the temperature settings, or the exact prompt. Furthermore, the 'hand-curated negative-surface list' for alias resolution is not provided. This lack of transparency prevents verification of the schema's integrity and potential bias in method identification.

paper_reviewer_figure_critic

Verdict: minor_revision

The review of figures for "Intern-Atlas" is currently constrained by the absence of the actual image files in the provided asset list. While the LaTeX source references five distinct figures (Figures 1–5), the file list only contains logos and a few specific PDFs ('figures/Figure3.pdf', 'figures/exp01_static_scores.pdf', etc.), but notably lacks 'intro.pdf', 'method.pdf', and 'figures/exp02_overall_tier_scatter.pdf' (referenced as 'exp02_overall_tier_scatter.pdf' in the code but potentially missing or misnamed in the asset list).

Specific concerns regarding the available figures: 1. **Missing Primary Visuals**: Figures 1 and 2 are critical for understanding the "methodological evolution graph" concept and the system pipeline. Without 'intro.pdf' and 'method.pdf', it is impossible to evaluate the clarity of the graph topology, the legibility of the node labels (e.g., "Transformer", "BERT"), or the effectiveness of the color coding for edge types (extends, improves, etc.). 2. **Legibility and Labels**: For Figure 4 (main_four_results), the scatter plot component (Figure 4d) must have clearly labeled axes. The caption mentions "Overall scores" and "strata," but the visual must explicitly label the y-axis with the score range and the x-axis with the stratum categories. Similarly, Figure 5 (downstream-results) contains a heatmap; ensure the color map is distinct and the correlation values are readable. 3. **Alt Text**: The LaTeX source does not include '\alttext' or equivalent accessibility metadata for any figure. Given the paper's focus on AI agents and structured data, ensuring these figures have descriptive alt text is crucial for accessibility and future machine parsing. 4. **Print Scale**: The pipeline figure (Figure 2) is complex, containing three distinct stages. Ensure that the text within the diagram (e.g., "Step 1: entity resolution") is large enough to be read when the figure is scaled down to a single column in the final PDF.

Please provide the missing figure files and verify that all axes, legends, and labels are

explicitly defined in the final compiled PDF.

- [writing] Figure 1 (intro.pdf) and Figure 2 (method.pdf) are referenced but the source files are missing from the provided asset list. The review cannot verify axis labels, color choices, or legibility for these primary figures. Please ensure all referenced PDFs are included in the submission.
- [writing] Figure 3 (Figure3.pdf) caption describes a 'detailed view of the LLM continent' but the file is listed as 'figures/Figure3.pdf'. The caption text is dense; ensure the visual legend or color coding for the 'six paradigm continents' is distinct enough to be legible at print scale without relying solely on the caption text.
- [writing] Figure 4 (main_four_results) combines a bar chart (a) and a scatter plot (d) in a single caption. The scatter plot (exp02_overall_tier_scatter.pdf) lacks visible axis labels in the filename description; verify that the final PDF includes explicit axis titles (e.g., 'Overall Score', 'Publication Stratum') and units where applicable, as the current LaTeX source does not show the plot generation code.
- [writing] Figure 5 (downstream-results) includes a heatmap (exp02_judgment_spearman_heatmap.pdf). Heatmaps require high-contrast color maps for print legibility. Verify that the color scale is colorblind-friendly and that the correlation values (Spearman coefficients) are legible within the cells or clearly indicated in a legend, as the current source does not show the plotting code.

paper_reviewer_jargon_police

Verdict: full_revision

The manuscript suffers from significant jargon overuse, frequently employing technical terms from graph theory, information retrieval, and deep learning without definition or simplification. This creates a barrier for non-specialist readers, including researchers in adjacent fields who might benefit from the infrastructure.

Specific instances include the use of 'SGT-MCTS' in the Abstract (line 14) and 'MCTS' in the Introduction (line 45) without defining the acronym first. The term 'parametric memory' (line 32) is used to describe an LLM's internal state, which is unnecessarily technical; 'internal knowledge' would suffice. Similarly, 'topological signals' (line 102) and 'causal topology' (line 48) rely on graph-theory jargon that could be replaced with 'structural indicators' and 'causal structure' respectively.

The acronym 'DAG' appears in the Introduction (line 52) without expansion. 'RAG' is used in the Experiment section (line 234) without definition. 'BM25' is introduced in the Method section (line 118) without context for those unfamiliar with information retrieval. The phrase 'lossy compression' (line 33) is a technical term that could be simplified to 'imperfect summary' for broader accessibility.

Furthermore, terms like 'verbatim source evidence' (line 18) and 'methodological evolution topologies' (line 34) are dense. While 'verbatim' is precise, 'direct quotes' is more

accessible. 'Topologies' in this context is redundant; 'structures' or 'patterns' would be clearer.

The paper assumes a high level of familiarity with specific AI research infrastructure concepts (e.g., 'strong-causal subset', 'alias resolution', 'stub nodes') without providing a glossary or simplified explanations. This exclusionary language contradicts the paper's stated goal of serving as a foundational layer for a broader ecosystem of automated discovery. To improve accessibility, the authors must define all acronyms at first use and replace domain-specific jargon with plain English equivalents where possible.

- [science] The manuscript suffers from significant jargon overuse, frequently employing technical terms from graph theory, information retrieval, and deep learning without definition or simplification. This creates a barrier for non-specialist readers, including researchers in adjacent fields who might benefit from the infrastructure. Specific instances include the use of 'SGT-MCTS' in the Abstract (line 14) and 'MCTS' in the Introduction (line 45) without defining the acronym first. The term 'parametric m

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a compelling architecture for a methodological evolution graph, but several logical gaps exist between the stated premises and the derived conclusions.

First, the evaluation of the SGT-MCTS algorithm (Table 1, Sec 4.1) claims a 39.9 point improvement in Node Recall over Beam@10. While the arithmetic (84.8 - 44.9) is correct, the causal explanation—that MCTS's exploration of "under-visited branches" leads to higher recall than a beam search that retains top-k paths—is not rigorously supported. In dense graphs, beam search typically maximizes recall by keeping multiple high-probability paths. The paper asserts that MCTS finds "more" nodes, but does not logically explain why the beam search failed to find them if the edge confidences (the heuristic) were accurate enough to guide the beam. If the heuristic was poor, the graph construction itself is suspect. The link between the algorithm's "exploration" mechanism and the specific metric gain is asserted rather than derived from the graph's topology.

Second, there is a contradiction regarding the "deterministic" nature of the Idea Evaluator. The main text (Sec 2.2) emphasizes that the system avoids LLM bias by using "deterministic graph statistics" and has "zero trainable parameters." However, Appendix A.4.3 describes an "Optional Adjudicator" that uses an LLM to rate ideas and set upper bounds on scores. If this component is enabled (as implied by its inclusion in the system description), the system is no longer purely deterministic or parameter-free. The conclusion that the system provides "full auditability" is logically weakened by the presence of a non-deterministic, black-box veto mechanism that can alter the final score.

Finally, the argument that Intern-Atlas solves the "structural understanding" limitation of LLMs is only partially supported. While the graph provides the *motivation* (bottleneck) for a new idea, the actual *technical proposal* (the mechanism) is still generated by an

LLM (Sec 2.3, Eq 5). The paper claims this prevents “fabricating proposals out of nothing,” but it does not logically demonstrate how the graph prevents the LLM from hallucinating the **feasibility** or **validity** of the proposed mechanism. The LLM still performs the “lossy compression” of generating the technical details, which the introduction claims is the root cause of the problem. The solution addresses the **search** for the gap but not the **generation** of the solution’s core logic, leaving a logical gap in the claim of a complete solution.

- [writing] The paper presents a compelling architecture for a methodological evolution graph, but several logical gaps exist between the stated premises and the derived conclusions. First, the evaluation of the SGT-MCTS algorithm (Table 1, Sec 4.1) claims a 39.9 point improvement in Node Recall over Beam@10. While the arithmetic (84.8 - 44.9) is correct, the causal explanation—that MCTS’s exploration of “under-visited branches” leads to higher recall than a beam search that retains top-k paths—is not rigor

paper_reviewer_overreach

Verdict: full_revision

The paper significantly over-claims the reliability and causal certainty of its core artifact, the Intern-Atlas graph. The Abstract and Introduction repeatedly assert that the graph is “grounded in verbatim source evidence” and forms a “queryable causal network.” However, Appendix A.3 reveals that the edge typing accuracy for the production model is only 70.4%, with the audit model reaching 93.0%. Given that the graph contains 9.4 million edges, a 30% error rate in the primary extraction phase means millions of edges are likely misclassified or contain hallucinated bottleneck evidence. Claiming “verbatim” grounding for a system with such a high error rate is scientifically unsound; the text should be revised to reflect that evidence is “LLM-extracted and heuristically validated” rather than strictly grounded.

Furthermore, the evaluation of the Idea Generation module (Section 4.3) over-attributes performance gains to the graph structure. The system uses the graph to identify gaps but relies on an LLM to generate the actual proposal text. The reported 88% win rate against a “No-KB” baseline does not sufficiently isolate the graph’s contribution from the LLM’s parametric knowledge of the specific research queries. Without a controlled ablation where the LLM is given the same structural hints via a different mechanism (or a strict control on the LLM’s context window), the claim that the **graph** is the primary driver of improved idea quality is an over-extrapolation.

Finally, the Conclusion states that the system “recovers expert-curated evolution chains more faithfully.” Table 1 shows a Node Recall of 84.8% and Edge Recall of 79.0%. While these are improvements over baselines, they indicate that nearly 20% of the ground-truth lineage is missed. Describing this as “faithful” recovery without explicitly acknowledging the ~20% failure rate in the main text is an over-interpretation of the results. The authors must temper their language to accurately reflect the probabilistic nature of the reconstruction rather than implying near-perfect fidelity.

- [science] The claim of “verbatim source evidence” for 9.4M edges is over-claimed given the 70.4% Phase-1 extraction accuracy (Appendix). A 30% error rate invalidates the “causal” certainty implied in the Abstract.
- [science] The Idea Generation evaluation (Sec 4.3) over-attributes win-rate gains to the graph. The LLM generates the text; the graph only identifies gaps. The study fails to isolate the graph’s contribution from the LLM’s parametric knowledge.
- [writing] The Conclusion claims “faithful” recovery of expert chains, yet Table 1 shows ~15-20% Node/Edge Recall loss. Describing this as faithful without quantifying the error rate in the main text is an over-interpretation.

paper_reviewer_safety_ethics

Verdict: minor_revision

The paper presents a significant infrastructure for automated scientific discovery, but several safety and ethical considerations require clarification before acceptance.

****Data Licensing and Copyright Compliance**** The methodology relies on extracting “verbatim quotes” (Eq. 1, Sec 3.2) from 1,030,314 papers spanning 1965–2025. While arXiv preprints are generally open, the corpus includes major conferences (NeurIPS, ICML, CVPR, etc.) where full texts are often behind paywalls or subject to strict copyright transfer agreements. The paper must explicitly state the licensing status of the full-text corpus used for extraction. Specifically, does the release of the graph and the verbatim evidence records comply with the copyright terms of the publishers for the non-arXiv papers? Without this clarification, the public release of the dataset poses a significant legal and ethical risk.

****Privacy and Consent in Evaluation Data**** The evaluation of the idea generator and evaluator utilizes a “Strata Dataset” (Sec 4.2) containing 300 rejected submissions from ICLR 2026. The use of rejected papers, which are often not publicly archived in the same manner as accepted works, raises concerns regarding author consent and privacy. Did the authors of these rejected papers consent to their work being used to train or evaluate an automated system that might critique or generate competing ideas? If not, this constitutes a potential violation of academic norms regarding the treatment of unpublished or rejected work. The paper should clarify the provenance and consent status of this specific subset of data.

****Bias Amplification and Dual-Use Risks**** The “Broader Impact” section (App E) acknowledges the risk of amplifying citation biases but offers only generic mitigation (verbatim grounding). Given that the system is designed to generate new research ideas (Sec 3.3), there is a risk that it could systematically favor “safe,” high-probability ideas derived from dominant paradigms, further marginalizing novel but less-cited approaches. Furthermore, the ability to automatically generate and evaluate research ideas could be misused to flood the scientific literature with low-quality or adversarial content. The authors should expand the Broader Impact section to include specific strategies for detecting and mitigating bias amplification and potential misuse in automated idea generation.

- [writing] The paper claims to process 1M+ papers from 1965-2025 (Sec 2.2, App C). Explicitly state the data licensing status for all venues (e.g., arXiv vs. paywalled proceedings). Confirm that the 'verbatim quote' extraction (Eq 1) complies with fair use and copyright restrictions for the specific venues included.
- [science] The 'Strata Dataset' (Sec 4.2) includes rejected submissions from ICLR 2026. Clarify the consent and anonymization protocol for these rejected papers. If authors were not explicitly consented for this specific evaluation, the use of their rejected work for training/evaluating an automated idea generator raises ethical concerns regarding academic privacy and potential bias against rejected work.
- [writing] The 'Idea Generation' operator (Sec 3.3) proposes new research ideas based on 'structural gaps.' Discuss the risk of the system amplifying existing citation biases (e.g., favoring well-cited methods) or generating ideas that inadvertently infringe on ongoing, unpublished work by the community. A 'Broader Impact' section is present but lacks specific mitigation strategies for these dual-use risks.

paper_reviewer_scientific_evidence

Verdict: full_revision

The scientific evidence supporting the core claims of Intern-Atlas is currently insufficient to warrant acceptance. While the system architecture is sophisticated, the empirical validation relies heavily on proxies and lacks rigorous statistical grounding.

First, the evaluation of the SGT-MCTS algorithm (Section 4.1, Table 1) presents point estimates (e.g., Node Recall 84.8%) without any measure of variance, confidence intervals, or statistical significance testing. Given that MCTS is a stochastic algorithm and the benchmark is derived from survey papers (which may have subjective interpretations of "evolution"), the observed improvements over Beam Search could be due to random variance or specific benchmark artifacts rather than a robust algorithmic advantage. The authors must report results over multiple random seeds and provide statistical tests (e.g., t-tests or bootstrap confidence intervals) to substantiate the claim of "much closer" recovery.

Second, the evaluation of the idea evaluator (Section 4.2) suffers from a fundamental circularity. The system is trained on a corpus of published papers, and its performance is validated against a "Strata Dataset" where the ground truth is the *publication venue* of the paper. Since the graph is constructed from these same venues, the system is effectively being tested on whether it can recognize the patterns of the data it was built from. This does not prove the system can evaluate the *quality* of an idea; it only proves it can identify papers that look like previously accepted papers. The claim that the graph provides a "more reliable basis" than LLMs is not supported by an independent ground truth (e.g., future citation impact or blind expert evaluation of the *idea* stripped of its publication context).

Third, the human evaluation for idea generation (Section 4.3) reports high win rates (88% vs No-KB) but lacks necessary statistical details. The sample size (100 queries) and

the specific blinding protocol are not fully transparent. If experts could infer the source of the idea based on writing style or specific formatting, the "blind" nature of the evaluation is compromised. Additionally, the comparison against BM25 RAG (81% win rate) is less dramatic, suggesting the advantage may be marginal in realistic retrieval scenarios.

Finally, the reliance on LLMs for the core graph construction (edge typing, bottleneck extraction) introduces a significant source of noise. The paper acknowledges a 70.4% accuracy for the production model in edge classification but fails to perform an error propagation analysis. It is unclear how a 30% error rate in identifying causal relationships impacts the downstream lineage reconstruction and idea generation. Without quantifying the sensitivity of the final results to these extraction errors, the robustness of the "causal network" claim remains unproven. The authors must address these methodological gaps by providing statistical rigor, independent ground-truth evaluations, and error analysis.

- [science] The scientific evidence supporting the core claims of Intern-Atlas is currently insufficient to warrant acceptance. While the system architecture is sophisticated, the empirical validation relies heavily on proxies and lacks rigorous statistical grounding. First, the evaluation of the SGT-MCTS algorithm (Section 4.1, Table 1) presents point estimates (e.g., Node Recall 84.8%) without any measure of variance, confidence intervals, or statistical significance testing. Given that MCTS is a stochastic

paper_reviewer_statistical_analysis

Verdict: full_revision

The statistical analysis presented in the paper is insufficient to support the strong claims made regarding the performance of Intern-Atlas. While the system architecture is sophisticated, the empirical validation lacks rigorous statistical grounding.

First, the results in Table 1 (Section 4.1) regarding the SGT-MCTS algorithm are highly suspicious. The reported Node Recall (NR) and Chain Alignment Score (CAS) are identical to one decimal place for every method (e.g., Beam@10: 44.9 for both). Since NR measures the fraction of nodes recovered and CAS measures the preservation of ordering, these metrics should theoretically diverge, especially for baselines like Random Walk which recover nodes but likely fail on ordering. This identity suggests a potential calculation error or a copy-paste mistake in the manuscript, rendering the claimed 39.9-point improvement over Beam@10 unverified.

Second, the human evaluation studies (Sections 4.2 and 4.3) rely on point estimates (means and correlations) without any measure of uncertainty. For the idea evaluation alignment (N=100), the paper reports Spearman correlations of 0.81 vs. 0.58 but provides no confidence intervals or p-values to test if the difference is statistically significant. Similarly, the idea generation scores in Table 3 are presented as means without standard deviations. Without error bars or significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests), it is impossible to determine if the observed improvements are robust or artifacts of sampling noise.

Finally, the analysis of the 'Strata Dataset' (Table 2) asserts a monotonic trend across publication tiers but does not provide a statistical test for trend (such as the Jonckheere-Terpstra test) or an ANOVA to confirm that the variance between groups is significantly larger than the variance within groups. The lack of these standard statistical controls makes the conclusions about the system's ability to stratify quality premature. The authors must re-run the analyses to include confidence intervals, p-values, and effect sizes, and correct the apparent data reporting error in Table 1.

- [science] The statistical analysis presented in the paper is insufficient to support the strong claims made regarding the performance of Intern-Atlas. While the system architecture is sophisticated, the empirical validation lacks rigorous statistical grounding. First, the results in Table 1 (Section 4.1) regarding the SGT-MCTS algorithm are highly suspicious. The reported Node Recall (NR) and Chain Alignment Score (CAS) are identical to one decimal place for every method (e.g., Beam@10: 44.9 for both). Si

paper_reviewer_text_formatting

Verdict: minor_revision

The manuscript demonstrates a high level of technical sophistication, but several text formatting inconsistencies require attention to ensure a clean compilation and professional presentation.

First, there is a discrepancy in **Figure 4** (Section 4.1). The caption explicitly describes subfigure (b) as "Distribution of Overall scores across strata," yet the LaTeX code only defines subfigures labeled (a) and (d). The label `\label{fig:four_d}` is assigned to the second subfigure, but the caption text refers to it as (b). This inconsistency between the visual labels, the caption text, and the code structure must be resolved to avoid reader confusion.

Second, the **bibliography** section presents formatting risks. The `ref.bib` file contains duplicate `@String` definitions (e.g., `PAMI`, `CVPR` are defined twice with slightly different expansions). While LaTeX may not error out, this is poor hygiene and can lead to inconsistent abbreviation rendering. Additionally, the `unsrt` bibliography style is used, but the entry types vary inconsistently (e.g., some conference papers use `booktitle` while others use `journal`), which may result in non-uniform citation formatting.

Third, in **Appendix A.1**, the "Bottleneck dimension taxonomy" table is initially commented out (lines 630-650) but an active version follows. The active table uses the `tabularx` environment with an `X` column for definitions. Given the length of some operational definitions, there is a risk of awkward line wrapping or overfull boxes if the column width is not perfectly tuned. It is recommended to verify the rendering or consider `longtable` if the content is dense.

Finally, a minor path issue exists in **Figure 3**: the code references `figures/Figure3.pdf`, but the provided asset list shows `figures/figure3.pdf` (case sensitivity or path depth should be double-checked against the actual file system to prevent compilation failures).

Addressing these formatting details will ensure the paper compiles without warnings and presents a polished appearance.

- [writing] In Section 4.1 (exp01), the caption for Figure 4 references subfigure (b) ('Distribution of Overall scores...'), but the LaTeX code only defines subfigures (a) and (d). The caption text and labels are inconsistent with the provided code structure.
- [writing] The bibliography style 'unsrt' is used, but the .bib file contains duplicate @String definitions (e.g., 'PAMI', 'CVPR') and inconsistent formatting for conference proceedings (some use 'booktitle', others 'journal'). This will cause compilation warnings and inconsistent citation rendering.
- [writing] In Appendix A.1, the 'Bottleneck dimension taxonomy' table (tab:dim-taxonomy) is commented out in the source code (lines 630-650) but referenced in the text. The active table (lines 652-670) uses 'tabularx' which may cause line-breaking issues in the 'X' column if the definitions are too long; ensure the column width is sufficient or switch to 'longtable' if it spans pages.
- [writing] Figure 3 caption references 'Figure3.pdf' but the file path in the code is 'figures/Figure3.pdf'. Ensure the file path matches the actual directory structure to avoid 'File not found' errors during compilation.

paper_reviewer_writing_quality

Verdict: minor_revision

The paper demonstrates a high level of technical sophistication and presents a compelling narrative for the need for methodological evolution graphs. However, the writing quality requires minor revisions to ensure clarity, grammatical correctness, and smooth flow.

****Clarity and Grammar:**** There are several instances of missing spaces after periods (e.g., Introduction, Section 1) and incomplete sentences (e.g., "Idea evaluation places a research idea" in Section 3.2.2). These typos, while minor, detract from the professional polish of the manuscript. Additionally, some sentences are overly dense and could benefit from breaking into shorter, more digestible units to improve readability.

****Flow and Cohesion:**** The transition between paragraphs in the Related Work section (Section 2) is occasionally abrupt. For instance, the shift from discussing OpenAlex to introducing Intern-Atlas could be smoother with a clearer bridging sentence. Similarly, the introduction of the SGT-MCTS algorithm in Section 3 feels slightly disconnected from the preceding overview; a brief sentence explaining **why** MCTS is chosen over other methods would enhance the logical flow.

****Consistency:**** There are minor inconsistencies in figure labeling and caption references (e.g., Figure 3 vs. Figure3.pdf). Ensuring uniformity in these details is crucial for a polished final product.

****Recommendation:**** The authors should carefully proofread the manuscript to address the identified typos and grammatical issues. Additionally, revising the transitions

between sections and paragraphs will improve the overall coherence and readability of the paper. These changes are straightforward and do not require re-running experiments or altering the core scientific content.

- [writing] In Section 1 (Introduction), the sentence 'Their parametric memory is a lossy compression that underrepresents low-frequency or long-tail methodological knowledge.Their autoregressive inference...' is missing a space after the period. Please correct this typo.
- [writing] In Section 2 (Related Work), the phrase 'Intern-Atlas bridges this critical infrastructure gap' appears in the middle of a paragraph without a preceding sentence break or clear transition from the previous discussion of OpenAlex. Consider adding a transition sentence or breaking the paragraph for better flow.
- [writing] In Section 3 (Method), the phrase 'Idea evaluation places a research idea' in the first sentence of Section 3.2.2 is grammatically incomplete and unclear. It likely means 'Idea evaluation places a research idea within the methodological landscape.' Please rephrase for clarity.
- [writing] In Section 4 (Experiment), the caption for Figure 3 (Figure~ ef{fig:intern-atlas-overview-half}) contains a minor inconsistency: it refers to 'Figure 3' in the text but the label is 'Figure3.pdf'. Ensure consistency between figure labels and references in the text.
- [writing] In Appendix A (Graph Construction), the table caption for Table~ ef{tab:dim-taxonomy} is repeated twice in the source code (once commented out, once active). Remove the commented-out version to avoid confusion during compilation.